Luke Cook

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https://www.linkedin.com/in/invocatis	https://www.github.com/invocatis
Bachelor's[Computer Science] - 2017	Rochester Institute of Technology

Senior Software Engineer (2022-2024) **Software Engineer (2020-2022)**	December 2020 - July 2024
Datasembly	**Remote**

Responsibilities
- Develop and maintain automated data collection systems for grocery-related datasets
- Reverse engineer proprietary web APIs and integrate them into a unified data pipeline
- Design and implement evasion techniques to ensure high data acquisition success rates
- Collaborate with Product and Customer teams to transform raw data into actionable insights

Projects
- Led development of a Playwright-based browser automation framework, simplifying integration logic and supporting advanced features such as JavaScript rendering, API encryption handling, and dynamic login workflows
- Initiated effort to define and standardize the 'Out-of-Stock' detection logic, collaborating with cross-functional teams to align on data parsing patterns. Improved understanding, and clarity into data meaning.

Software Engineer	March 2020 - September 2020
Hyperfiddle	**Remote**

Responsibilities
- Built full-stack features in a Clojure/ClojureScript + Datomic application framework
- Implemented CI/CD pipeline using CircleCI, AWS ECR/ECS, and Terraform
- Built configurable sample data generation library to support product demos and internal testing workflows

Projects
- Designed an extensible algebra system for Datomic statements
  - Enabled automatic simplification of negating statements (e.g., 1 + -1 = 0)
  - Developed frontend logic to emit statements that self-simplify on change
  - Proved Datomic statements formed an Algebraic Group structure

Application Programmer II	June 2019 - June 2020
Cornell University	**Ithaca, NY**

Responsibilities

- Redesigned genomic CSV ingestion tool for flexibility and maintainability
  - Simplified instruction file format (V2), reducing redundancy and improving readability
  - Rewrote core digester to use declarative Aspect Files (V3), which described the structure of file
  - Developed a web app to convert instruction files between V1, V2, and V3 formats in Clojure(script)

Projects

- Built an automated end-to-end testing framework for genomic data pipelines
  - Designed a domain-specific language to structure tests using Arrange/Act, Assert, and Cleanup phases
  - Implemented architecture with a Clojure backend, Java logic layer, and custom scripting syntax
  - Ensured test independence and idempotency to support safe execution in production-like environments

Application Programmer II	June 2019 - June 2020
Cornell University	**Ithaca New York**

Responsibilities

- Improved genomic CSV digester application
  - Simplified instruction fiel format, reducing redundancy (V2)
  - Rewrote digester to be more generic and powerful, developed declaritive Aspect Files (v32)
  - Created a file translator to convert instruction files between V1, V2, and V3 in Clojure(script)

Projects

- Developed framework for automated end-to-end testing
  - Created a domain-specific scripting langauge for test structure: Arrange/Act, Assert, Cleanup
  - Clojure for backend, Java for business logic, and custom syntax for test specifics
  - Ensured test independnence and idempotency in production environments

Software Engineer Co-Op	Jan 2018 - July 2018
Intuit	**Mountain View, CA**

Responsibilities

- Contributed to an agile team developing microservices for account creation workflows
- Implemented a document upload endpoint for identity verification during onboarding
- Increased test reliability and confidence by expanding unit test coverage across critical components

Skills

Programming Languages

Scala, Clojure(script), Java, SQL, Rust, Python, Javascript

Infrastructure

PostgreSQL, Snowflake, Airflow, MySQL, Kafka, Linux

Devops

Docker, Terraform, GCP, AWS, CircleCI, Bamboo

Monitoring	**Theory**
Grafana, Prometheus, BugSplat, Slack Integration	Functional Programming, Object Oriented Programming, Type Theory, Language & Framework Design, Typed Functional Programming

Projects

eldritch *Algebraic Data Types & Pattern Matching* https://www.github.com/invocatis/eldritch	**motif** *Recursive Pattern Matching in Clojure* https://www.github.com/invocatis/motif
ernie *Testing Framework for Java written in Clojure* https://www.github.com/invocatis/ernie	**Scala 3 Algebraic Type Exploration** *Exploration of algebraic concepts defined in Scala 3* https://github.com/Invocatis/scala3-algebraic-type-exploration
CSV Digester *CSV ingestion and manipulation library* https://bitbucket.org/gobiiproject/gobii.masticator/src/develop/	